

Reliable Evaluation of ARIMA-LSTM Hybrids for Remaining Useful Life Prediction: A Multi-Seed, Specification-Sensitive Study on NASA C-MAPSS

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Abstract: A common design choice for remaining useful life (RUL) prediction is to augment an LSTM with a classical trend feature, typically ARIMA, fit per sensor channel. We ask whether this holds up under multi-seed, multi-specification, statistically corrected evaluation on NASA C-MAPSS, rather than proposing it as a new architecture. We compare a per-channel ARIMA-LSTM hybrid against a plain LSTM, Random Forest, and CNN-LSTM on all four C-MAPSS sub-datasets under six seeds, testing three ARIMA specifications: a fixed order, an AIC-selected order, and a fully causal expanding-window construction that never uses future information. Under the causal, deployable specification, the hybrid shows no Holm-Bonferroni-corrected advantage over plain LSTM on any dataset (closest case FD002, adjusted $p = 0.058$). A decisive control replacing ARIMA with a free, fitting-free causal moving average significantly *outperforms* causal ARIMA on FD002 (adjusted $p = 0.023$) at one-quarter the compute cost. Under the specification used in most prior work, a fixed ARIMA order is significantly worse than plain LSTM on FD004 ($p = 0.004$); AIC-based order selection repairs that disadvantage without beating plain LSTM anywhere. No specification tested yields a corrected-significant hybrid-over-plain win. The contribution is methodological: evaluation choices, seeds, specification, and causal validity, determine what a predictive-maintenance study can defensibly claim.

Index Terms: remaining useful life, predictive maintenance, LSTM, time-series forecasting, feature engineering, industrial IoT, NASA C-MAPSS, reproducibility

I. INTRODUCTION

Estimating remaining useful life (RUL) from streaming sensor data is central to prognostics and health management (PHM), letting maintenance be scheduled proactively rather than reactively [1]. Industrial IoT deployments produce continuous, multivariate sensor telemetry, vibration, temperature, pressure, rotational speed, from equipment such as turbines, pumps, and CNC machinery, and a recent survey [10] catalogs the resulting breadth of deep-learning approaches: CNNs, autoencoders, GANs, transformers, and foundation models, across supervised, transfer, and federated learning paradigms. NASA’s C-MAPSS turbofan simulation [1] is the field’s standard benchmark: a controlled, multi-regime, multi-fault-mode setting where model-comparison claims can in principle be checked. A common architectural idea is to fit an ARIMA

model per sensor channel and concatenate its trend with the raw signal before an LSTM regressor [17], letting the network spend capacity on residual nonlinear dynamics rather than re-deriving a smooth trend from scratch.

But deep models are trained stochastically, and an ARIMA feature is itself a specification choice, order, fitting window, causality with respect to the prediction target, not a fixed quantity. A single-seed, single-specification claim that a trend feature helps cannot be trusted at face value; prior turbofan RUL papers, including hybrid-forecasting work we cite favorably [9], typically report neither seed count nor causal validity. This paper evaluates, rather than proposes, ARIMA trend augmentation for LSTM-based RUL prediction. We test the hybrid against plain LSTM, Random Forest, and CNN-LSTM baselines across all four C-MAPSS sub-datasets, six seeds each, under three ARIMA specifications, a fixed order matching prior work’s common default, a per-channel AIC-selected order, and a fully causal expanding-window construction, applying Holm-Bonferroni correction consistently across every four-dataset inferential family.

Research questions. RQ1: does a fully causal ARIMA trend feature consistently improve LSTM RUL prediction across the four C-MAPSS subsets? RQ2: how sensitive are plain-vs-hybrid comparisons to stochastic training variation? RQ3: does ARIMA order specification materially change the outcome under the protocol used in most prior work? RQ4: can a much cheaper, fitting-free causal trend summary reproduce whatever effect ARIMA has? RQ5: what methodological conclusions survive once every comparison is corrected?

Contributions: (1) a multi-seed, four-dataset evaluation of ARIMA-LSTM RUL prediction against matched plain-LSTM, CNN-LSTM, and Random Forest baselines; (2) a causal feature-construction analysis removing a train/test asymmetry present in the standard full-series ARIMA fit; (3) an order-specification sensitivity analysis, explicitly labeled as evidence under the original non-causal protocol rather than validated causal evidence; (4) a decisive control showing a free causal moving-average feature significantly beats causal ARIMA on the one dataset ARIMA looks most promising on;

and (5) consistent Holm-Bonferroni correction across every inferential family, backed by publicly released code and peer-reviewed results (full experimental record, including the window-length ablation and gated-fusion check, at <https://github.com/PradeepSomasundaram1512/hybrid-arima-lstm-rul-cmapss>).

II. RELATED WORK

RUL estimation on C-MAPSS. The C-MAPSS benchmark [1] underlies most data-driven RUL work; Ramasso and Saxena [2] laid out early benchmarking protocols. Recent architectures include CNN-LSTM [5], residual-corrected BiLSTM [6], and attention-based BiLSTM [8] hybrids; Su and Fang [7] decompose degradation into trend, operational-state, and residual components via a learned embedding, a philosophy that motivates our own trend/residual split, though implemented with ARIMA. We do not compare our numbers against these directly, since protocols (RUL cap, window length, train/validation split) may differ and an uncontrolled cross-paper comparison is not defensible either way; the point of this paper is precisely that such comparisons require matched protocol to mean anything.

Multi-condition RUL estimation. FD002 and FD004 introduce six discrete operating regimes and, in FD004’s case, two concurrent fault modes, both increasing estimation difficulty relative to FD001’s single-condition setting. Su and Fang [7] decompose degradation into trend, operational-state embedding, and residual-noise components via a learned embedding, a philosophy that motivates our own trend/residual split, though implemented with ARIMA rather than a learned representation.

Hybrid statistical/neural forecasting. ARIMA-neural hybrids have a long history in general forecasting [17] and have been applied to RUL-adjacent estimation problems [9], typically without reporting seed count or the causal validity of the ARIMA construction, both questions this paper treats as central rather than incidental.

Reproducibility and seed sensitivity. The instability we document (Section V) is a specific instance of a broader problem elsewhere: single-seed deep reinforcement learning results routinely fail to replicate [11], and exact reproducibility does not imply robustness to small setup changes [12], a point a companion paper [13] operationalizes with concrete variance-estimation protocols. Individual random seeds can differ by margins comparable to the gap between competing published methods [14], a variance traced to a calibration property of the trained network rather than unstructured noise [15]. Demšar [16] sets out the paired, multiple-comparison-corrected testing methodology we follow. A contemporaneous benchmark on this exact dataset [18] uses a single fixed seed, and a parallel audit [19] independently finds preprocessing choices materially affect reported C-MAPSS metrics, a concern parallel to ours though focused on regime normalization rather than trend-feature specification. Our approach differs from prior turbofan RUL work in using ARIMA purely as a per-channel feature-engineering step, an explicit trend covariate per sensor, rather than a separate forecaster fused post-

hoc with a deep model’s prediction, and in treating ARIMA specification and causal construction as first-class experimental variables, evaluated across all four regimes, six seeds, and three specification strategies, rather than a single convenient run reported without any of these checks.

III. METHODOLOGY

Data and preprocessing. We use all four C-MAPSS sub-datasets [1]: FD001/FD003 (100 engines, single/dual fault mode, 1 condition), FD002/FD004 (260/249 engines, 1/2 fault modes, 6 conditions). RUL is capped at 125 cycles; low-variance sensors are dropped (15 of 21 retained for FD001/FD003). For FD002/FD004, we K -means cluster the three operating-setting dimensions into 6 regimes (fit on training data only) and z -score normalize per regime using regime-specific training statistics [4], clipped to ± 6 std; the partition is confirmed seed-stable up to label permutation (100% agreement after relabeling).

ARIMA trend feature, three specifications. For each (unit, sensor), we fit ARIMA and use its in-sample fitted trend $\hat{y}_t$ as an extra input channel, concatenated with raw features before a 2-layer LSTM (hidden size 64) whose cell updates follow the standard formulation [3]:

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad h_t = o_t \odot \tanh(c_t), \quad (1)$$

followed by a $64 \rightarrow 32 \rightarrow 1$ regression head. *Fixed order* (main comparison): ARIMA(1, 1, 0), parameters fit once via MLE over each unit’s full available series. *AIC-selected order*: the best of four candidate orders, (1, 1, 0), (0, 1, 1), (1, 1, 1), (2, 1, 0), under the identical full-series fit, isolating order choice from causality; we label this explicitly as a specification-sensitivity analysis under the non-causal protocol, not causal evidence. *Causal, expanding-window*: ARIMA(1, 1, 0) refit by full MLE every 20 cycles on an expanding window, with Kalman-filter state extension between refits, so no parameter is ever estimated from data at or after the cycle it predicts; this is our primary, deployable specification.

Baselines and protocol. A plain LSTM (identical architecture, raw features only) isolates the trend feature’s effect; a Random Forest (300 trees, max depth 10) is trained on per-cycle flat snapshots; a CNN-LSTM (2-layer 1D-CNN front end, no ARIMA feature) [5] tests whether any seed-sensitivity finding is specific to the LSTM backbone. All models use Adam (10^{-3}), MSE loss on capped RUL, 15% held-out validation engines for early stopping (patience 8, max 60 epochs), window length 30 cycles. Each test engine is scored once from its final window. We report RMSE, MAE, and the asymmetric PHM score [1]: writing $d_i = \widehat{\text{RUL}}_i - \text{RUL}_i$,

$$s_i = \begin{cases} \exp(-d_i/13) - 1, & d_i < 0 \\ \exp(d_i/10) - 1, & d_i \geq 0, \end{cases} \quad (2)$$

penalizing late (over-optimistic) predictions more heavily than early ones.

TABLE I

LSTM HYPERPARAMETERS (IDENTICAL FOR PLAIN AND HYBRID; INPUT WIDTH DIFFERS ONLY IN TREND-CHANNEL COUNT).

Hyperparameter	Value
LSTM layers / hidden size	2 / 64
Optimizer / loss	Adam, 10^{-3} / MSE on capped RUL
Batch size / max epochs	128 / 60
Early-stopping patience	8 epochs
Window length / RUL cap	30 cycles / 125 cycles
Validation split	15% of training engines
Seeds (all specifications)	42, 123, 2024, 7, 99, 314

TABLE II

DATASET SCALE AND APPROXIMATE SINGLE-SEED TRAINING RUNTIME.

Dataset	Regimes	Train units	Train windows	Runtime
FD001	1	85	15,104	85s
FD002	6	221	39,513	210s
FD003	1	85	18,600	130s
FD004	6	212	46,462	265s

IV. EXPERIMENTAL SETUP

All runs use six seeds (42, 123, 2024, 7, 99, 314), identically applied to every specification and baseline; PyTorch 2.14.0 (CPU-only, Apple M2 Pro), statsmodels 0.15.0, and scikit-learn 1.9.1. Table I summarizes hyperparameters. We started with three seeds, found a significant result, and only treated it as reliable after replicating with three more, which changed the answer (Section V-B). Holm-Bonferroni correction is applied within every four-dataset inferential family in this paper. We do not enable PyTorch’s deterministic-algorithms mode, so bit-exact reproduction of a given seed’s weights is not guaranteed across hardware, only approximate reproduction of the reported metrics. The causal ARIMA construction logs its fallback rate explicitly: 0 of 165,216 series, across all 24 causal-specification runs, fell back to a rolling-mean approximation.

V. RESULTS

A. RQ1: Causal, Deployable Hybrid vs. Plain LSTM

Table III compares plain LSTM against the causal-ARIMA hybrid. The pattern is inconsistent across regimes: worse than plain on FD001 (15.39 vs. 14.43) and FD004 (15.59 vs. 14.92), close on FD002 (14.64 vs. 14.59), better on FD003 (14.22 vs. 14.72). A paired Wilcoxon test on per-engine error, Holm-corrected across the four datasets, finds no significant difference on any dataset (closest: FD002, raw $p = 0.0145$, adjusted $p = 0.058$; against the non-causal hybrid, closest is FD003, raw $p = 0.0172$, adjusted $p = 0.069$). *RQ1 answer*: no consistent improvement, and no dataset survives correction. The regime-by-regime pattern, worse on the two single-condition-adjacent extremes and better on FD003, is inconsistent rather than systematic, and we do not have a mechanistic account for it.

TABLE III

FIXED-ORDER HYBRID RMSE, CAUSAL VS. NON-CAUSAL ARIMA, MEAN $\pm$ STD OVER 6 SEEDS.

Dataset	Plain	Non-causal hybrid	Causal hybrid
FD001	14.43 $\pm$ 0.64	14.92 $\pm$ 1.65	15.39 $\pm$ 1.11
FD002	14.59 $\pm$ 0.49	14.93 $\pm$ 0.60	14.64 $\pm$ 0.36
FD003	14.72 $\pm$ 2.06	14.87 $\pm$ 0.58	14.22$\pm$0.56
FD004	14.92 $\pm$ 0.38	15.71 $\pm$ 0.54	15.59 $\pm$ 0.44

TABLE IV

RMSE ACROSS NASA C-MAPSS, MEAN $\pm$ STD OVER 6 SEEDS, NON-CAUSAL PROTOCOL.

Dataset	Plain LSTM	Hybrid ARIMA-LSTM	CNN-LSTM
FD001	14.43$\pm$0.64	14.92 $\pm$ 1.65	14.47 $\pm$ 0.59
FD002	14.59$\pm$0.49	14.93 $\pm$ 0.60	15.11 $\pm$ 1.05
FD003	14.72 $\pm$ 2.06	14.87 $\pm$ 0.58	13.90$\pm$0.70
FD004	14.92$\pm$0.38	15.71 $\pm$ 0.54	14.86 $\pm$ 0.54

B. RQ2: Sensitivity to Stochastic Training Variation

Under the original (non-causal) protocol, Table IV shows the plain LSTM’s mean RMSE at or below the hybrid’s on every dataset. This comparison itself moved with seed count: at three seeds, a paired Wilcoxon test found FD001 significant ($p = 0.007$) and nowhere else, and we had drafted an explanation for that specific result; adding three more seeds erased FD001’s significance ($p = 0.157$) and made FD004 significant instead ($p = 0.004$, Holm-adjusted, the only survivor across the four-dataset family). Cohen’s d_z and bootstrap 95% CIs confirm only FD004’s interval excludes zero ($d_z = 0.15$, CI [0.11, 1.01]); every effect is small by convention. A CNN-LSTM baseline (no ARIMA feature) shows the same dataset-dependent pattern rather than a uniform result: significantly worse than plain on FD002 ($p < 0.001$), non-significantly best-in-table on FD003 ($p = 0.091$), within noise on FD001/FD004. *RQ2 answer*: which specific dataset a significance claim can defend changed materially between three and six seeds, before any specification question was even asked; the overall direction (hybrid mean RMSE at or above plain on every dataset) held up both times, but which dataset a defensible significance claim could single out did not.

C. RQ3: Order-Specification Sensitivity (Non-Causal Protocol)

Holding the full-series fitting procedure fixed and varying only ARIMA order changes the FD004 picture materially. Order selection overwhelmingly preferred (0, 1, 1) over the fixed default (75–83% of unit/sensor/seed combinations). Table V reports Holm-corrected significance at two aggregation levels: seed-level ($n=6$) and per-engine ($n=100$ –259). Under either design, exactly one cell is significant: selected order beats *fixed* order on FD004 (engine-level, $p_{\text{adj}} = 0.0004$), repairing the fixed-order hybrid’s RQ2 disadvantage. Nothing survives correction for selected order vs. *plain* LSTM on any dataset; FD002’s largest numerical gap (13.77 vs. 14.59 RMSE) does not survive ($p_{\text{adj}} = 0.125$ seed-level). We label this explicitly

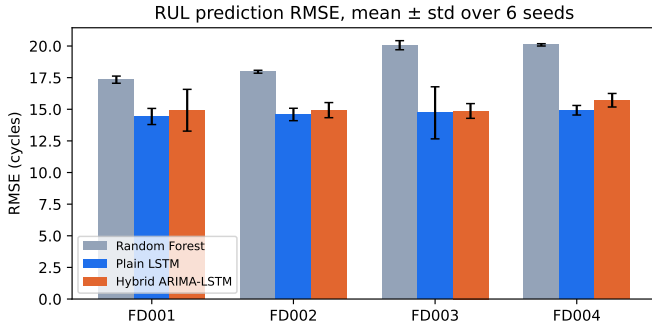


Fig. 1. RMSE across all four sub-datasets, mean $\pm$ 1 std over six seeds, non-causal protocol. Plain LSTM’s mean RMSE is at or below the hybrid’s on every dataset; note FD003’s wide error bars, driven mostly by the plain model’s own across-seed variance there.

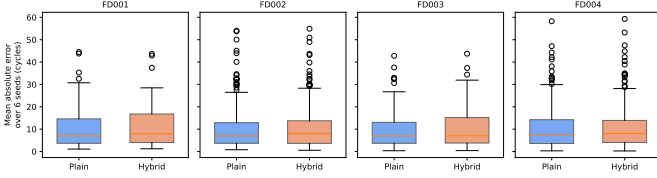


Fig. 2. Per-engine absolute error, averaged over six seeds, plain vs. non-causal hybrid LSTM, across all four datasets.

as a specification-sensitivity finding under the original, non-causal protocol, not evidence that a causally-fit, AIC-selected model would behave the same way, a question we have not tested.

D. RQ4: Can Cheaper Causal Smoothers Reproduce ARIMA’s Effect?

We replace the causal-ARIMA feature with a trailing 10-cycle moving average (MA) and an exponential moving average (EMA, $\alpha = 0.3$), both fitting-free and trivially causal, under the identical six-seed protocol (Table VI). Neither MA nor EMA differs significantly from plain LSTM on any dataset (Holm-adjusted $p \geq 0.91$). Against causal ARIMA, however, MA is significantly *better* on FD002 (raw $p = 0.0058$, Holm-adjusted $p = 0.023$): a mean per-engine error reduction of 0.629 cycles ($d_z = -0.14$, 95% CI $[-1.16, -0.07]$), at one-quarter the wall-clock cost (501s vs. 2103s mean per-seed runtime). *RQ4 answer*: a free causal smoother matches or beats causal ARIMA on the dataset that matters most, arguing against ARIMA’s specific autoregressive structure rather than trend augmentation broadly.

E. RQ5: Additional Robustness Checks and Synthesis

A window-length ablation on FD001 (six seeds, $W \in \{10, 20, 30, 50\}$, Table VII) shows the non-causal hybrid’s advantage concentrated at $W=20$ and $W=30$ (0.68 and 0.54 cycles) and within noise at the extremes (-0.02 at $W=10$, -0.17 at $W=50$), not the clean peak-and-reverse pattern an earlier three-seed version suggested.

A gated-fusion variant, a learned per-timestep sigmoid gate on the AIC-selected trend feature, our direct attempt at an

TABLE V
SELECTED- VS. FIXED-ORDER AND SELECTED- VS. PLAIN-LSTM, HOLM-ADJUSTED p -VALUES, NON-CAUSAL PROTOCOL.

Comparison (adj. p)	FD001	FD002	FD003	FD004
vs. fixed, seed-level	0.688	0.124	0.438	0.124
vs. fixed, engine-level	0.516	0.086	0.086	0.0004
vs. plain, seed-level	1.000	0.125	1.000	1.000
vs. plain, engine-level	0.849	0.849	0.257	0.114

TABLE VI
CAUSAL TREND-FEATURE CONTROLS, MEAN $\pm$ STD OVER 6 SEEDS.

Dataset	Plain	MA-hybrid	Causal ARIMA
FD001	14.43 $\pm$ 0.64	14.93 $\pm$ 0.86	15.39 $\pm$ 1.11
FD002	14.59 $\pm$ 0.49	14.10$\pm$0.45	14.64 $\pm$ 0.36
FD003	14.72 $\pm$ 2.06	14.87 $\pm$ 0.87	14.22$\pm$0.56
FD004	14.92 $\pm$ 0.38	14.81$\pm$0.45	15.59 $\pm$ 0.44

architectural fix for a poorly-specified trend feature, looked promising at a single-seed pilot (competitive with the selected-order hybrid on three of four datasets) but shows no significant difference from plain LSTM or the selected-order hybrid at full six-seed power (Table VIII, all $p > 0.13$). A mechanistic check asks whether FD002 has more ARIMA-explainable sensor trends than FD004, which would explain its numerical gap: fitting ARIMA(0, 1, 1) to 25 training units per dataset and measuring the fraction of each differenced sensor’s variance the fitted trend explains, a naive per-sensor test (treating each of 364–500 correlated unit-sensor pairs as independent) originally suggested FD002’s mean R^2 was higher than FD004’s, $p < 0.05$. Sensors within a unit share that unit’s degradation trajectory and are not independent, so we recomputed at the unit level (25 unit-mean values per dataset, the correct statistical unit); the FD002-vs-FD004 gap is no longer significant ($+0.007$, $p = 0.107$). We do not have a supported mechanistic account for FD002’s numerical gap, and report that as an open question rather than an asserted one. Full tables for all three checks are in the repository.

Across every specification tested and after consistent correction, we find exactly two corrected-significant effects in this paper, and neither is a hybrid-over-plain win: AIC-selected order beats fixed order on FD004 (RQ3), and a free moving average beats causal ARIMA on FD002 (RQ4). *RQ5 answer*: no comparison of any ARIMA-augmented hybrid against plain LSTM survives correction, under any specification, on any dataset; a naively fixed order actively hurts (RQ2), correcting it repairs that harm without producing a win (RQ3), and a free trend summary matches or beats the causal construction where it matters most (RQ4).

VI. DISCUSSION

What we think is actually going on. Under the original protocol, ARIMA is fit to each unit’s full cycle-indexed series, not to the 30-cycle window the LSTM consumes. An ARIMA(1, 1, 0) model is restrictive, a single autoregressive lag on the first difference, and our order-selection results show

TABLE VII
WINDOW-LENGTH ABLATION ON FD001, NON-CAUSAL HYBRID, MEAN $\pm$ STD OVER 6 SEEDS (RMSE, CYCLES).

W (cycles)	10	20	30	50
Plain	18.60 $\pm$ 0.55	17.39 $\pm$ 0.41	14.69 $\pm$ 0.68	14.46 $\pm$ 0.69
Hybrid	18.62 $\pm$ 0.65	16.71 $\pm$ 0.29	14.15 $\pm$ 0.70	14.63 $\pm$ 1.01

TABLE VIII
GATED-FUSION RMSE, MEAN $\pm$ STD OVER 6 SEEDS, VS. PRIOR MODELS (NON-CAUSAL PROTOCOL).

Dataset	Plain	Fixed	Selected	Gated
FD001	14.43	14.92	14.38	14.97 $\pm$ 0.74
FD002	14.59	14.93	13.77	13.93 $\pm$ 0.35
FD003	14.72	14.87	14.45	14.33 $\pm$ 0.90
FD004	14.92	15.71	14.90	15.03 $\pm$ 0.47

that assumption was wrong 75–83% of the time, $(0, 1, 1)$ fit better almost everywhere. A trend feature built from a poorly specified model is not neutral, it is actively misleading, consistent with the fixed-vs-selected comparison in RQ3, where correcting the order specifically repairs FD004’s disadvantage. But removing the deeper train/test asymmetry via the causal construction (RQ1) does not, on its own, turn any dataset into a corrected-significant win, which tells us specification quality and causal validity are separate axes of the same evaluation problem, not substitutes for each other.

The paper’s value does not depend on ARIMA winning. A naively fixed order significantly hurts on FD004; correcting the order repairs that specific harm without producing a win; removing the deeper causal asymmetry (RQ1) does not turn any dataset into a corrected-significant win either; and where a trend feature shows its largest promise (FD002), a free rolling mean beats the expensive ARIMA construction outright (RQ4). The order-selection frequency is itself seed-stable (0.0 percentage-point std within each dataset), so the specification-sensitivity finding is not a training-stochasticity artifact: the ARIMA order-selection step is deterministic given the data, and whatever varies between seeds comes from the LSTM’s own training stochasticity, not order instability. Taken together, specification quality and causal validity are separate axes of evaluation, not substitutes for one another: a single-seed or single-specification claim on this benchmark, or similar small-to-moderate industrial datasets, is not sufficient evidence either way, and comparison against a cheap non-ARIMA baseline is cheap enough to be routine.

Is FD002’s numerical gap concentrated in one operating regime, or broad? Broad, with one exception, though this describes a non-corrected-significant effect (RQ3), not a further hypothesis test. Stratified by FD002’s six operating regimes, the order-selected hybrid’s mean error is lower than the plain LSTM’s in five of six regimes (0.3–1.4 cycles lower MAE) and higher in one (regime 5, by 1.1 cycles), an exception we cannot explain. FD003’s plain-model across-seed std (2.06 RMSE, Table IV) is separately the largest of any model-dataset pair

reported here; we have no confident explanation and would rather flag it as unexplained than invent one.

What we would tell someone deciding whether to add this kind of feature. Do not treat a single run, or even three seeds, as sufficient evidence either way; this paper’s own headline claim changed twice as seeds were added and, separately, once more as ARIMA specification and causal construction were varied. Four checks are cheap enough to be routine on this benchmark: rerun under five or six seeds, not one to three; correct for multiple comparisons whenever a test repeats across datasets; treat a fixed ARIMA order as a choice to be checked, not assumed; and, if a feature is meant to be deployed rather than only evaluated post hoc, verify its construction is actually causal with respect to the prediction target, since a disclosed but uncorrected asymmetry is not the same thing as a validated deployable method.

VII. LIMITATIONS

Six seeds is a matched, moderate-power design, not exhaustive stochastic validation, though sufficient here to overturn this paper’s own three-seed conclusion twice; the RQ2 reversal (FD001’s significance disappearing while FD004’s emerged) is a direct illustration that even six seeds should be read as a lower bound on the evidence needed, not a settled sample size chosen in advance. All experiments use simulated turbofan data; generalization to real industrial IoT sensor streams with different noise and missingness characteristics is untested. The causal specification costs $3.7\text{--}10.1\times$ the non-causal pipeline’s runtime, a practical deployment consideration we surface but do not optimize, and this cost itself scales with dataset size and CPU contention rather than being a fixed constant. We do not have an explanation for FD003’s unusually high plain-model variance across seeds (Table IV), nor for why FD002’s operating regime 5 runs against the pattern seen in its other five regimes; a fuller account would need per-sensor rather than per-dataset analysis, which we did not attempt. The window-length ablation uses a shorter training budget than the main experiments and has not been rerun under the main protocol’s exact budget. We tested three ARIMA specifications, one recurrent architecture, and two fusion mechanisms (concatenation and gating), but not an auxiliary trend-prediction loss or a sweep of LSTM hidden size or depth, either of which could plausibly interact with the specification-sensitivity effect reported in RQ3.

VIII. CONCLUSION

We evaluated, rather than proposed, whether giving an LSTM an explicit per-sensor ARIMA trend feature improves remaining useful life prediction on NASA C-MAPSS, treating specification and causal construction as first-class experimental variables alongside architecture. Evaluated across three ARIMA specifications, four C-MAPSS sub-datasets, and six seeds each with consistent Holm-Bonferroni correction, an ARIMA-augmented LSTM shows no corrected-significant advantage over a plain LSTM anywhere, under its causal, deployable construction or otherwise. Order specification and causal

validity each materially change individual comparisons without changing this overall conclusion, and a free causal moving-average control significantly outperforms causal ARIMA on the one dataset the hybrid looked most promising on. The methodological point, that evaluation choices, seeds, specification, and causal validity, determine what a predictive-maintenance study can defensibly claim, is the paper’s contribution, independent of whether this specific hybrid is adopted, and we expect it generalizes to other statistical/neural hybrids evaluated on small-to-moderate benchmark datasets. Future work includes a mechanistic account of why FD002’s numerical gap is largest, an auxiliary trend-prediction loss as an untested alternative to gating, a window-length ablation rerun under the main experiments’ exact training protocol, and validating any of these specifications on real, non-simulated industrial IoT sensor data.

CODE AND DATA AVAILABILITY

Full code, per-seed results, the complete technical report (all research questions, ablations, and negative results reported here in full), and this paper are available at <https://github.com/PradeepSomasundaram1512/hybrid-arima-lstm-rul-cmapss>: 150 pipeline executions across all specifications and baselines, training 360 individual models in total.

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